Assessment Brief

Module Leader: Mehmet Ozcan		Level: 5	
Module Name: Algorithms and Data Structures		Module Code: 55-508810	
Assignment Title: Portfolio			
Individual	Weighting: 100%	Magnitude: 5000 words	
Submission date/time: 7th January 2025 by 3 pm	Blackboard submission: Yes Turnitin submission: Yes	Format: Word, source code, digital media.	
Planned feedback date: 28th January 2025	Mode of feedback: Written	In-module retrieval available: No	
In this assessment are students asked to consider:	Inclusivity and accessability		Not applicable
	Sustainability		Not applicable

Module Learning Outcomes

- LO1: Design algorithms in an appropriate abstract notation and characterise them and describe their attributes using appropriate metrics.
- LO2: Compare and contrast similar algorithms and describe their relative merits and demerits.
- LO3: Design novel algorithms.
- LO4: Optimise implementations of algorithms.
- LO5: Construct and manipulate the complex data structures required of many algorithms using appropriate programming languages.

This portfolio represents **100% of the module**. It involves individual work. It addresses all the learning outcomes of the module

A portfolio consists of a set of projects. A project is essentially a series of tasks based on specific topics taught throughout the semester. Each project will contribute differently to the overall module mark, deliver specific artefacts, test knowledge and understanding of topics as well as their application to solve computer science problems. A portfolio is an individual work. The number, nature, difficulty level and marking weight of projects will vary from year to year. A portfolio includes the following artefacts (note that not all these artefacts may be needed for each project):

1. a set of algorithms written in an algorithmic language (e.g., Algorithmic Definition Language or ADL for short).
2. a set of data structures represented as diagrams to store data.
3. a piece of software written as a console application in C#[1].
4. a video presentation that showcases the capabilities of software in (2).

[1] If you did not learn C# in your first year, then you must approach us.

5. a descriptive report on any aspect of design and development, including artefacts produced for each project.
6. a short report outlining response to formative feedback.

These will be elaborated next.

Algorithms

You are expected to express your solution to a given problem using an algorithmic language, such as ADL taught in the lecture programme. Your algorithm should not bias towards a particular programming language, such as C#, nor should it be expressed in terms of it. Where possible, your algorithm should be described in terms of suitable abstractions, i.e., either procedural or functional or in terms of classes. You are encouraged to consider the notion of complexity while devising an algorithm and make judicious choices if there are alternative solutions to a given problem.

Data Structures

You are expected to define, describe, and justify data structures needed to manipulate data used in an application. You are also expected to illustrate the data structures with example data. You are encouraged to consider alternative data structures for a given problem and make judicious choices using the notion of space-speed trade off.

Software and its Video Presentation, including testing

You are expected to submit each project, including all its components (e.g., codebase), compressed in a separate zip file (or 7z), which should be named as Project 1, Project 2, and Project 3, etc. Your zip files must not contain shortcuts. The zip files will be uploaded to Blackboard as directed in the relevant submission point on Blackboard. We may at our discretion request a demonstration if we have difficulty in understanding or running your software.

Each project needs to be showcased in a separate video recording of up to 10 minutes. We will stop watching after the 10th minute. There must be sound in your recording. We will stop watching if your video resembles a Charlie Chaplin film. You are expected to upload your video files to Panopto and include their links in your portfolio. Your video files should be named as Project 1 Video, Project 2 Video and Project 3 Video, etc. The following format is advised:

- Introduce yourself.
- Introduce the problem.
- Demonstrate the functionality of your software using different data sets.
- Explain how you tested your software or tested your results/findings (if necessary).

Where applicable, each project should contain a light-touch test specification outlining how acceptance tests will be carried out as well as providing evidence of acceptance testing and its results. This should be provided in writing in portfolio.

Descriptive Report, including artefacts

In this section, you are required to elaborate on how you devised your algorithm and developed the corresponding implementation in C#. In software development, several other artefacts in addition to algorithms and data structures are also produced. **Each software project has its own requirements and deliverables. The following may be required (specific requirements will be outlined in project specifications)**:

- Description of how you transformed your algorithm into a corresponding implementation in C#.
- Time-Space Trade-off of algorithms used.
- Computational complexity of algorithms used.
- Problem-solving strategy.
- Judgement on results and personal opinions
- Analysis of results
- Tabulation of results
- Drawing a line graph for results

Incorporation of formative feedback (typically 250 words)

For each project, you are expected to provide written evidence of how you evaluated and acted on the formative feedback you received from your tutors. Written evidence can include in what form you received feedback (e.g., written via email or verbal in a tutorial or support session), who gave you feedback and when (e.g., a specific date). Note that formative feedback will mainly be given verbally in tutorials. In addition, you are entitled to email us only once per project to seek feedback. You are welcome to show us your progress and receive further feedback in tutorials as already stated. **Do not fabricate feedback to avoid allegation for academic misconduct.**

Assessment Criteria

Refer to the University Grade Descriptor, which describes the relationship of degree classification to grade point and equivalent percentage.

	FAIL (insufficient)				THIRD (sufficient)			LOWER SECOND (good)			UPPER SECOND (very good)			FIRST (excellent)				
	Zero	Low Fail	Mid Fail	Marginal Fail	Low 3rd	Mid 3rd	High 3rd	Low 2.2	Mid 2.2	High 2.2	Low 2.1	Mid 2.1	High 2.1	Low 1st	Mid 1st	High 1st	Exceptional 1st	Perfect 1st
Criteria and weighting[2]	**<19**		**20-39**		**40-49**			**50-59**			**60–69**			**70-84**		**85+**		
C1: An assessment of time-space trade-off choices **Artefacts**: A set of data structures and corresponding data (where applicable) Constructive elaboration on the given problems in analysing the time-space trade-off choices.	Some attempt has been made to tackle the task. The work has little or no merit.		Work is not fit for purpose. There may be evidence of some basic understanding of relevant concepts, but this does not meet the level that was taught in class.		You show some awareness of the trade-off between time and space when you make your arguments. You show satisfactory awareness of the notion of a data structure. However, your choice of data structures could be better to achieve the required trade-off between time and space.			Competent work, demonstrating reasonable knowledge and understanding. Some analysis and appropriate skills are shown. The work as presented has some minor errors but is limited in scope.			High quality work, demonstrating good knowledge, understanding and analysis. The work as presented has some very minor errors, and there are some obvious areas where it could be improved/extended.			Very high-quality work, demonstrating excellent knowledge, understanding and analysis. The work as presented is difficult to fault, but there are some obvious areas where it could be extended/improved.		Exceptional work of the highest quality, demonstrating excellent knowledge, understanding and analysis. There is extensive evidence of independent investigation, learning and thought.	
C2: An assessment of ability to work out complexity of an algorithm **Artefacts**: Computational complexity expressed in terms of Big-O. Implementation of given pseudocodes to programs.	Some attempt has been made to tackle the task. The work has little or no merit.		Work produced is not fit for purpose. There may be evidence of some basic understanding of relevant concepts and techniques, but this does not meet the level that was taught in class.		Work of limited quality, demonstrating some relevant knowledge and understanding. You show some awareness of the notion of frequency counts. However, their translation to complexity of an algorithm and representation using the Big-O notation is not consistently accurate.			Competent work, demonstrating reasonable knowledge and understanding, analysis and accuracy. The work as presented has some minor errors but is limited in scope.			High-quality work, demonstrating good knowledge and understanding, analysis and accuracy. The work as presented has some very minor errors, and there are some obvious areas where it could be improved/extended.			Very high-quality work, demonstrating excellent knowledge and understanding, analysis and accuracy. The work as presented is difficult to fault, but there are some obvious areas where it could be extended/improved.		Exceptional work of the highest quality, demonstrating excellent knowledge and understanding, analysis and accuracy. It is difficult to suggest ways that the work could be improved/extended. You show excellent awareness of the notion of frequency counts of a given algorithm. You can accurately translate frequency counts to complexity of an algorithm and represent it using the Big-O notation.	
C3: An assessment of understanding of algorithm design	Some attempt has been made to tackle the task.		Work produced is not fit for purpose.		Work demonstrates some relevant			Competent work, demonstrating reasonable knowledge and			High-quality work, demonstrating good			Very high-quality work, demonstrating excellent		Exceptional work of the highest quality, demonstrating excellent knowledge and understanding.		

[2] Not all these criteria will apply to each project. Furthermore, the weightings of criteria that are applicable to each project may vary. You need to refer to specific project specifications for distribution of marks to each criterion.

	FAIL (insufficient)				THIRD (sufficient)			LOWER SECOND (good)			UPPER SECOND (very good)			FIRST (excellent)				
	Zero	Low Fail	Mid Fail	Marginal Fail	Low 3rd	Mid 3rd	High 3rd	Low 2.2	Mid 2.2	High 2.2	Low 2.1	Mid 2.1	High 2.1	Low 1st	Mid 1st	High 1st	Exceptional 1st	Perfect 1st
Criteria and weighting[2]	**<19**		**20-39**		**40-49**			**50-59**			**60–69**			**70-84**		**85+**		
Artefacts: A set of algorithms expressed in ADL or something similar Constructive elaboration on the given problems into appropriate data structure.	The work has little or no merit.		There may be evidence of some basic understanding of relevant concepts and techniques, but this does not meet the level that was taught in class.		knowledge and understanding. Your explanation of your problem-solving strategy lacks clarity. Your algorithms are correct with respect to the specification, i.e., it accomplishes its task unambiguously and accurately. Your algorithms are accurate in the way they manipulate data stored in your chosen data structures.			understanding. However, even though it is satisfactory, your problem-solving strategy lacks clarity and could be more effective. The work as presented has some minor errors but is limited in scope.			knowledge and understanding. The work as presented has some very minor errors, and there are some obvious areas where it could be improved/extended.			knowledge and understanding. The work as presented is difficult to fault, but there are some obvious areas where it could be extended/improved.		It is difficult to suggest ways that the work could be improved/extended. You are able explain your problem-solving strategy accurately and clearly. Your problem-solving strategy is effective. There is a very close mapping between your strategy and your algorithm. Your algorithms are correct with respect to the specification, i.e., it accomplishes its task unambiguously and accurately. Your algorithms are accurate in the way they manipulate data stored in your chosen data structures.		
C4: Results analysis and discussion **Artefacts:** Tabulated experiment results. Analysis of statistical values of the results (if applicable). Elaboration of the results and conclusion of the finding. Summary the results for the problem	Some attempt has been made to tackle the task. The work has little or no merit.		Work produced is not fit for purpose. There may be evidence of some basic understanding of relevant concepts and techniques, but this does not meet the level that was taught in class.		Work demonstrates some analysis and discussion. Results analysis section, which entails experiments for problem solving, is present. Result for each experiment is presented in the report. Summary of the results for the problem is presented.			Competent work, demonstrating reasonable analysis and understanding. The work as presented has some minor errors but is limited in scope.			High-quality work, demonstrating good analysis and understanding. The work as presented has some very minor errors, and there are some obvious areas where it could be improved/extended.			Very high-quality work, demonstrating excellent analysis and discussion. The work as presented is difficult to fault, but there are some obvious areas where it could be extended/improved.		Exceptional work of the highest quality, demonstrating excellent analysis and discussion. It is difficult to suggest ways that the work could be improved/extended. Excellent presentation of experiment results in tabular format which enables empirical analysis to be carried out to justify algorithms/experiments. Evidence of good results are produced and comparable to the metric measurement used for the problem. Excellent critical summary of the result section for findings. Excellent use of graphs or charts to visualise results to support findings.		

	FAIL (insufficient)				THIRD (sufficient)			LOWER SECOND (good)			UPPER SECOND (very good)			FIRST (excellent)				
	Zero	Low Fail	Mid Fail	Marginal Fail	Low 3rd	Mid 3rd	High 3rd	Low 2.2	Mid 2.2	High 2.2	Low 2.1	Mid 2.1	High 2.1	Low 1st	Mid 1st	High 1st	Exceptional 1st	Perfect 1st
Criteria and weighting[2]	**<19**		**20-39**		**40-49**			**50-59**			**60–69**			**70-84**		**85+**		
																	Excellent discussion on how to improve current experiment.	
C5: An assessment of data structure design **Artefacts:** A set of data structures expressed in terms of diagrams and sample data sets (where applicable)	Some attempt has been made to tackle the task. The work has little or no merit.		Work produced is not fit for purpose. There may be evidence of some basic understanding of relevant concepts and techniques, but this does not meet the level that was taught in class.		Choice of data structures is mostly accurate. Alternative data structures are either not considered or not considered consistently. Example data sets to illustrate how data may be stored using your chosen data structures (where appropriate) are presented.			Competent work, demonstrating reasonable analysis and understanding. The work as presented has some minor errors but is limited in scope.			High-quality work, demonstrating good analysis and understanding. The work as presented has some very minor errors, and there are some obvious areas where it could be improved/extended.			Very high-quality work, demonstrating excellent analysis and discussion, and presentation. The work as presented is difficult to fault, but there are some obvious areas where it could be extended/improved.		Exceptional work of the highest quality, demonstrating excellent knowledge and understanding, and presentation. It is difficult to suggest ways that the work could be improved/extended. Your choice of data structures is accurate. You are able to produce a number of alternative data structures by either using data provided to you or by analysing given problem domains. You are able to provide example data sets to illustrate how data may be stored using your chosen data structures (where appropriate).	
C6: An assessment of overall software, including testing	Some attempt has been made to tackle the task. The work has little or no merit		Work produced is not fit for purpose. There may be evidence of some basic understanding of relevant concepts and techniques, but this does not meet the level that was taught in class.		Your software is mostly fit for purpose. However, there is either no evidence of following a systematic approach to translate your algorithms to corresponding C# programs, or evidence is not consistent. There is a clear and light-touch test specification in place for acceptance tests. There is a video recording which satisfies the expectations laid out adequately, if not completely.			Competent work, demonstrating reasonable analysis and understanding. The work as presented has some minor errors but is limited in scope.			High-quality work, demonstrating good analysis and understanding. The work as presented has some very minor errors, and there are some obvious areas where it could be improved/extended.			Very high-quality work, demonstrating excellent analysis and discussion, accuracy, and presentation. The work as presented is difficult to fault, but there are some obvious areas where it could be extended/improved.		Exceptional work of the highest quality, demonstrating excellent knowledge and understanding, accuracy, and presentation. Your software is fit for purpose. There is clear evidence that you followed a systematic approach to translate your algorithms to corresponding C# or Java programs. There is clear and light-touch test specification in place for acceptance tests. There is evidence of successful execution of the test specification and recording of its results. Video recordings satisfactorily	

	FAIL (insufficient)				THIRD (sufficient)			LOWER SECOND (good)			UPPER SECOND (very good)			FIRST (excellent)				
	Zero	Low Fail	Mid Fail	Marginal Fail	Low 3rd	Mid 3rd	High 3rd	Low 2.2	Mid 2.2	High 2.2	Low 2.1	Mid 2.1	High 2.1	Low 1st	Mid 1st	High 1st	Exceptional 1st	Perfect 1st
Criteria and weighting[2]	**<19**		**20-39**		**40-49**			**50-59**			**60–69**			**70-84**		**85+**		
																demonstrate the functionality of software. outline any assumptions you made while developing the software.		
C7: Incorporation of formative feedback.	Some attempt has been made to tackle the task. The work has little or no merit.		Some attempt was made to receive feedback, but no or little attempt was made to respond to it.		Some attempt was made to receive feedback. Feedback received was documented. There is evidence of an action plan to address feedback, even though not all feedback was addressed.			Adequate attempt was made to receive feedback. Feedback received was documented well. There is evidence of an action plan to address feedback. Some of the feedback was addressed.			Satisfactory attempt was made to receive feedback. Feedback received was documented well. There is evidence of an action plan to address feedback. Most of the feedback was addressed.			Excellent attempt was made to receive feedback. There is clear evidence of evaluation of feedback received (e.g., in the form of a “to-do-list” with priorities). There is also convincing evidence of successful execution of the action plan.		Excellent attempt was made to receive feedback. There is clear evidence of evaluation of feedback received (e.g., in the form of a “to-do-list” with priorities) and recording the action plan using a software tool, such as Trello. There is also convincing evidence of successful execution of the action plan.		

University Grade Descriptor (UGD)

Level 5 - Generic grade descriptor: relationship of degree classification and equivalent percentage

Class	Category	Mark range	%	General Characteristics
1st	Exceptional 1st	93 - 100	96	Exceptional breadth and depth of knowledge and understanding of the area of study, **significantly beyond what has been taught in all areas**; evidence of extensive and appropriate selection and critical evaluation/synthesis/analysis and of reading/research beyond the prescribed range, in both breadth and depth, to advance work/direct arguments; excellent communication; performance beyond expectation. The ability to make decisions and carry out tasks/processes with autonomy; excellent leadership skills in group contexts; creative flair; extremely well-developed problem-solving skills; the ability to carry out sustained critical reflection on practical work within the wider context of industry/workplace. Fully meets expectations set by the industry/employment context.
1st	High 1st	85 - 92	89	Outstanding/excellent knowledge and understanding of the area of study as the student is **typically able to go beyond what has been taught (particularly for a mid/high 1st**); evidence of extensive and appropriate selection and critical evaluation/synthesis/analysis of reading/research **beyond the prescribed range**, to advance work/direct arguments; excellent communication; performance deemed beyond expectation of the level. The ability to make decisions and carry out tasks/processes with autonomy; creative flair and the ability to (re)interpret predefined rules/conventions to select and justify individual working practice; highly developed problem-solving skills; accuracy and fluency; excellent command of skills appropriate to the task; the ability to reflect critically on practical work within the wider context of industry/workplace. Broadly meets expectations set by the industry/employment context.
	Mid 1st	78 - 84	81	
	Low 1st	70 - 77	74	
2.1	High 2.1	67 - 69	68	Very good knowledge and understanding of the area of study as the student is **typically able to relate facts/concepts together with some ability to apply to known/taught contexts**; evidence of appropriate selection and evaluation of reading/research, some beyond the prescribed range, may rely on set sources to advance work/direct arguments; demonstrates autonomy in approach to learning; strong communication skills. Broadly autonomous completion of practical tasks/processes; ability to adapt in response to change or unexpected experiences; technical/artistic decision making is highly developed; a clear command of the skills relevant to the task/process; ability to reflect on practical work and set future goals within the wider context of industry/workplace. Adherence to standards set by the industry/employment context.
	Mid 2.1	64 -66	65	
	Low 2.1	60 - 63	62	
2.2	High 2.2	57 - 59	58	Good knowledge and understanding of the area of **study balanced towards the descriptive rather than analytical**; evidence of appropriate selection and evaluation of reading/research but generally reliant on set sources to advance work/direct arguments; communication shows clarity, but structure may not always be coherent. A confident approach to practical tasks; a solid grasp of the related processes, tools, technology; creativity in the completion of the task; proficiency is demonstrated by an accurate and coordinated performance; tasks are completed with a good level of independent thought; some autonomy is evident; an ability to reflect on practical work and set future goals. General adherence to standards set by the industry/employment context.
	Mid 2.2	54 - 56	55	
	Low 2.2	50 - 53	52	
3rd	High 3rd	47 - 49	48	**Knowledge and understanding sufficient to deal with terminology, basic facts and concepts** but fails to make meaningful synthesis; some ability to select and evaluate reading/research however work may be more generally descriptive; strong reliance on available support set sources to advance work; arguments may be weak or poorly constructed; communication/presentation is generally competent but with some weaknesses. Competence in technical/artistic skills; tasks/processes are completed with a degree of proficiency and confidence; tasks are completed with a basic level of independent thought; effective judgements have been made; basic evaluation and analysis of performance in practical tasks is evident. Errors in workflow or completion of the task; general adherence to appropriate rules/conventions set by the industry/employment context.
	Mid 3rd	44 - 46	45	
	Low 3rd	40 - 43	40	
Fail	Borderline Fail	30 - 39	35	Insufficient knowledge and understanding of the subject and its underlying concepts; **some ability to evaluate given reading/research however work is more generally descriptive; naively follows or may ignore set material in development of work**; given brief may be only tangentially addressed or may ignore key aspects of the brief; communication shows limited clarity, poor presentation, structure may not be coherent. Practical tasks are attempted; skill displayed in some areas; there are a significant number of errors; a lack of proficiency in most areas; guidance may be needed to reproduce aspects of the task and/or apply learned skills. Tasks may be incomplete; failure to adhere to some of the rules/conventions set by the industry/employment context.
	Mid Fail	20 - 29	25	
	Low Fail	6-19	10	No evidence of knowledge or understanding of the subject; **no understanding of taught concepts, with facts being reproduced in a disjointed or decontextualised manner**; ignores set material in development of work; fails to address the requirements of the brief; lacks basic communication skills. A general level of incompetency in practical tasks; an evident lack of practice; set tasks are not completed; few or no skills relating to tasks are evident. No adherence to rules/conventions set by the industry/employment context.
Zero	Zero	0-5	0	Work not submitted, work of no merit, penalty in some misconduct cases.

Artificial Intelligence and Academic Integrity – AI&AI

It is important you do not use AI tools to generate an assignment and submit it as if it were your own work. Our regulations state:

Contract cheating/concerns over authorship: This form of misconduct involves another person (or artificial intelligence) creating the assignment which you then submit as your own. Examples of this sort of misconduct include: buying an assignment from an 'essay mill'/professional writer; submitting an assignment which you have downloaded from a file-sharing site; acquiring an essay from another student or family member and submitting it as your own; attempting to pass off work created by artificial intelligence as your own. These activities show a clear intention to deceive the marker and are treated as misconduct.

Further guidance is available here: https://blogs.shu.ac.uk/assessment4students/preparing-to-submit-work/#AI

Live Demonstration

In some cases, we may require live demonstration in person for the submitted work without any justification. Failure to attend a demonstration without an extenuating circumstance will result in being awarded zero marks for the whole coursework. Demonstrations will take place in the first teaching week of Semester 2.

Project 1 (15%)

Problem domain: Algorithmic Complexity

The objective of this project is to enable you to apply the concept of algorithmic complexity to given problems. You are required to do the following tasks:

1. Work out the computational complexity of a given algorithm.
2. Analyse the time complexity of two given mathematical functions through experiment.
3. Produce a short report outlining response to formative feedback.
4. Add your solutions to your portfolio using the portfolio template provided.

This project represents 15% of the total marks available in this module.

Note that the deadline for completion of this project is the same as the submission date of your portfolio.

Task 1

Consider the following algorithm fragment written in ADL and calculate its computational complexity with justification and represent it in Big-O notation, i.e., $O(n)$. Your answer must be justified using a polynomial expression, i.e. using a $f(n)$, as outlined in the relevant lecture.

```
for i ← 1 to n by 1 do
  for j ← 1 to i by 1 do
    for k ← 1 to j by 1 do
      print("What a great challenge!")
    end
  end
end
```

Task 2

Consider the two functions n^2 and $2^n/4$, which represent time complexity, for various values of n. Design a simple experiment to observe rate of growth of these functions for a given set of n values and analyse the results.

Grading Criteria

This work will be graded using the criteria given below.

Criteria	Marks
C2: Ability to work out the complexity of an algorithm	80
C4: Results analysis and discussion	15
C7: Incorporation of formative feedback	5

Project 2 (25%)

Problem Domain: A Mazing Problem

The objective of this project is to enable you to apply the concepts of data structure design and implementation. You are required to do the following tasks:

1. Design data structure(s) for the given problem domain.
2. Devise a set of algorithms using an algorithmic language, such as ADL, to manipulate the data structures designed in (1).
3. Implement the algorithms you devised in (2) systematically in C# as a console application and test your implementation.
4. A video presentation that showcases the capabilities of the software in (3).
5. A short descriptive report on any aspect of design and development, including artefacts produced for the project.
6. Produce a short report outlining response to formative feedback.
7. Add your solutions to your portfolio using the portfolio template provided.

This project represents 25% of the total marks available in this module.

Note that the deadline for completion of this project is the same as the submission date of your portfolio.

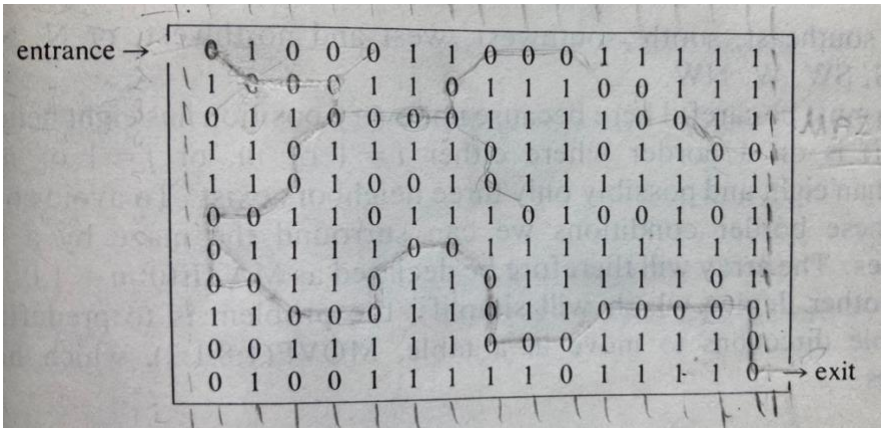
Rats are trained to run in a straight alley-like maze for a food reward which is located at the end of the alley.

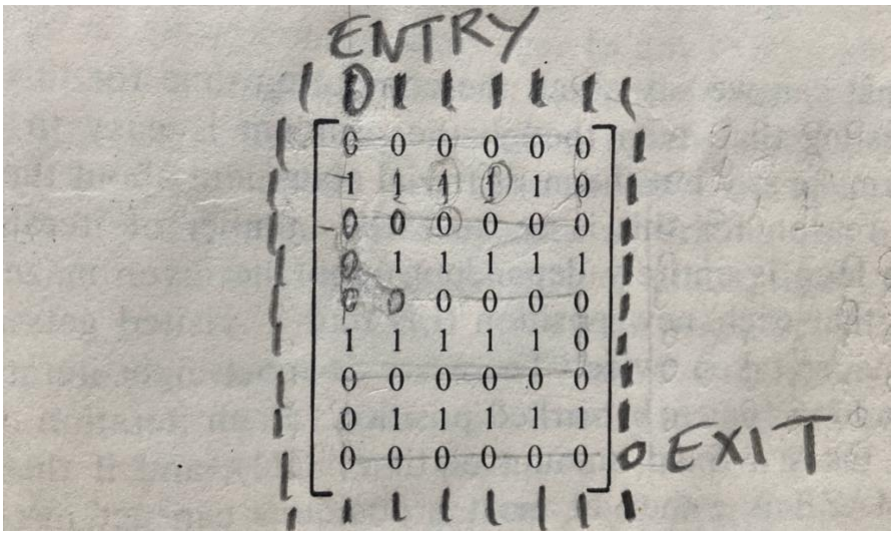
The rat-in-a-maze experiment is a classical one from experimental psychology. A rat (or mice) is placed through the door of a large box without a top. Walls are set up so that movements in most directions are observed. The rat makes its way through the maze while being carefully observed by scientists until it eventually reaches the exit at which it is rewarded with food. This experiment is run repeatedly until the rat will run through the maze without taking a single false path (i.e., a path that leads to a dead end). Through these trials, the learning curve of the rat is measured.

In this context, you are required to design and implement a simulation software for the “rat-in-a-maze” experiment. Input of the software is a maze which has the following properties:

1. it has one entry point and one exit point;
2. a rat can travel eight different directions, i.e., northwest, north, northeast, west, east, southwest, south and southeast;

Consider the following two maze examples in which “0” represents a path that is open and “1” represents a path that is closed, i.e., a wall. Note that in order to prevent a rat to escape from a maze, it is necessary to surround the outer perimeter of the maze with “1”s with the exception of the exit where a reward is offered to the rat. Note that once the rat enters the maze the entrance must be sealed.





Output of the software is a set of correct paths that enable a rat to reach exit. Note that if a maze does not have an exit (despite its aforementioned properties), then the program ought to display an appropriate message before aborting.

Grading Criteria

This work will be graded using the criteria given below.

Criteria	**Marks**
C3: An assessment of understanding of algorithm design	20
C5: An assessment of data structure design	30
C6: An assessment of overall software, including testing	45
C7: Incorporation of formative feedback	5

Project 3 (30%)

Problem Domain: Recognizing Key Influencers in Social Networks

The objective of this project is to enable you to apply the concepts of data structure design and implementation. You are required to do the following tasks:

1. Design data structure(s) for the given problem domain.
2. Devise a set of algorithms using an algorithmic language, such as ADL, to employ the data structures designed in (1).
3. Implement the algorithms devised in (2) systematically in C# as a console application and test the implementation.
4. A video presentation of not more than 10 minutes that showcases the capabilities of software in (3).
5. A short descriptive report of not more than 1200 words on
   a. The design and development of the problem
   b. An analysis of the data structure(s) selected for implementation, focusing on time and space efficiency.
   c. An evaluation of the implementation's scalability to the road network available at euroroad | Road Networks | Network Data Repository (networkrepository.com) where Edges represent roads and nodes are intersections where roads meet.
   d. A response to formative feedback, including the instructor's name and the timestamp.
6. Add your solutions to your portfolio using the portfolio template provided.

This project represents 30% of the total marks available in this module.

Note that the deadline for completion of this project is the same as the submission date of your portfolio.

Problem description

Given a social network modelled as a graph with users as nodes and connections as edges, the objective of this project is to identify users capable of achieving the widest reach across the entire network in the shortest time. This ability is quantified using a metric known as the ***influence_score***, which assesses how proximate a user is to all others within the network. I***nfluence_score*** is defined in terms of shortest paths (also known as geodesic distances) between nodes.

Problem statement:

Given a graph G=(V,E) where V is the set of nodes and E is the set of edges, the closeness score of node ***u*** is defined as the reciprocal of the sum of the shortest path distances from ***u*** to all other nodes in the graph. Formally, it can be expressed as given in Equation 1.

$$influence_score(u) = \frac{n-1}{\sum_{v \in V, u \neq v} d(u,v)} \qquad (1)$$

Where ***n*** is the number of nodes in G and ***d(u,v)*** is the shortest path distance between node ***u*** and node ***v***.

The computation of ***d(u,v)*** differs depending on whether the graph is weighted or unweighted.

- For unweighted graph, ***d(u,v)*** is the length of path with the fewest edges between ***u*** and ***v***.
- In a weighted graph, edges have different weights. The shortest path is the path with the minimum total edge weight between ***u*** and ***v***.

See appendix below for example illustrations.

Questions:

- Write an algorithm using ADL to calculate the ***influence_score*** of a node in unweighted network.
- Write an algorithm using ADL to calculate the ***influence_score*** of a node in weighted network.
- Implement these two algorithms in C#.

Submissions

1. Report
2. Code
3. Video

Grading Criteria

This project will be graded using the criteria given below.

Criteria	**Marks**
C1: An assessment of time-space trade-off choices	5
C2: An assessment of ability to work out complexity of an algorithm	5
C3: An assessment of understanding of algorithm design	10
C5: An assessment of data structure design	20
C6: An assessment of overall software, including testing	35
C7: Incorporation of formative feedback	25

Appendix: Example Illustration

Refer to the edge_list of an unweighted social network presented in Figure 1, given in List 1 as well as edge_list of a weighted social network presented in Figure 2, given in List 2.

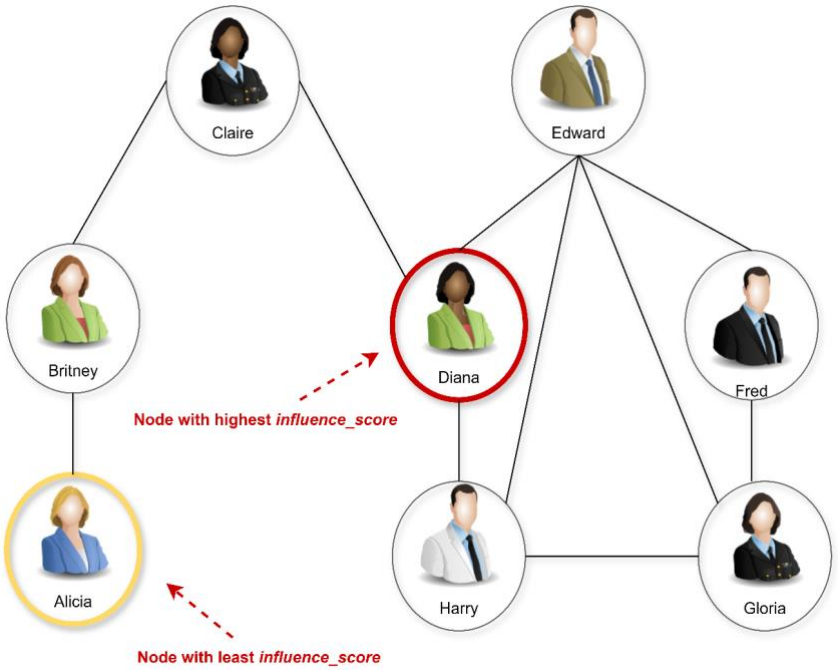


Figure 1 Unweighted Social Network

List 1: edge_list of unweighted social network given in Figure 1

Node1	Node2
Alicia	Britney
Britney	Claire
Claire	Diana
Diana	Edward
Diana	Harry
Edward	Harry
Edward	Gloria
Edward	Fred
Gloria	Fred
Harry	Gloria

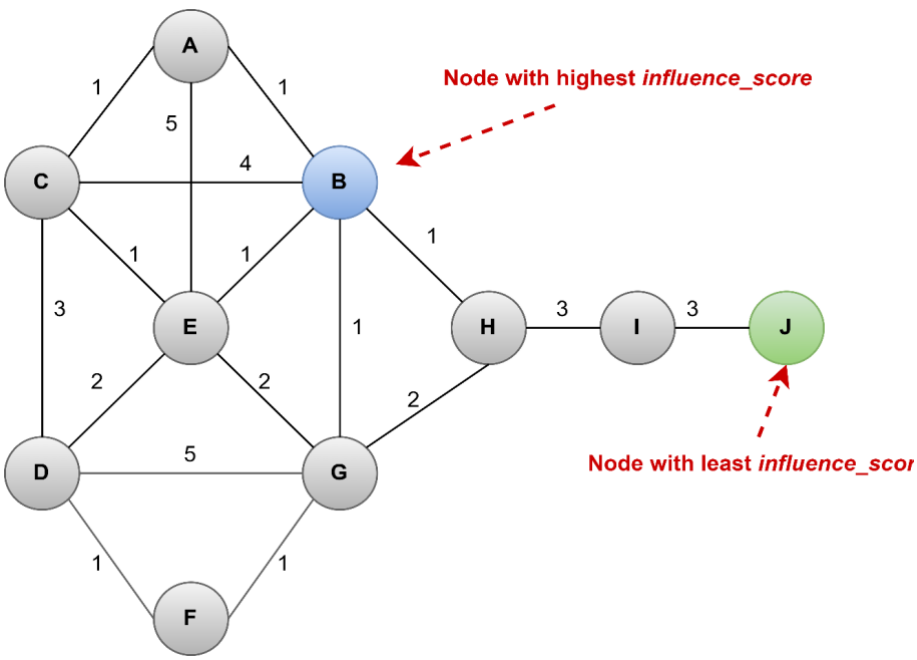


Figure 2 Weighted Social Network

List 2: edge_list of weighted social network given in Figure 2

Node1	Node2	Weight
A	B	1
A	C	1
A	E	5
B	C	4
B	E	1
B	G	1
B	H	1
C	D	3
C	E	1
D	E	2
D	F	1
D	G	5
E	G	2
F	G	1
G	H	2
H	I	3
I	J	3

The ***influence_score*** of Diana in Figure 1 would be 7÷(3+2+1+1+2+2+1) = 0.58, which is highest and that of Alicia is 7÷(1+2+3+4+5+5+4)= 0.29, which is least of the scores of all others.

In the weighted graph of Figure 2**,** the shortest path is not the one with the least hops but the one with the minimum total weighted distance. The ***influence_score*** of node B is 9÷(1+2+3+1+2+1+1+4+7) = 0.41, which turns out to be the node with the highest ***influence_score*** in this network, and node J with a score of 9÷(8+7+9+10+10+9+8+6+3) = 0.13, resulting to be the node with least ***influence_score***.

Project 4 (30%)

Solving The Bin Packing Problem using Hill Climbing

The objectives of this project are to comprehend and evaluate the given problem, to design a heuristic algorithm, to conduct empirical experiments as well as solving the problem. In this context, you are required to carry out the following tasks:

1. Conduct desk research on Bin Packing and document your understanding of the problem. Citations and references are required for this task.
2. Explain in writing for your understanding about the Random Mutation Hill Climbing method. Citations and references may be required (if applicable).
3. The requirements of Project 4 are loosely defined. You are responsible to seek advice from your tutors to progress with the project. Consistently get feedback from your tutor about the task and document them in the report.
4. Identify the best data structure to represent the solution.
5. Design and describe the algorithm for the problem.
6. Conduct empirical experiments to solve the problem.
7. Analyse results in the report and write recommendations from the experiment results.
8. Demonstrate your solution and its artefact (code-based) in a video recording.
9. Document your work in the report.

This project represents **30%** of the total marks available in this module.

Note that the deadline for completion of this project is the same as the submission date of your portfolio.

Desk Research: The Bin Packing Problem

The Bin Packing problem is a popular paradigm problem in heuristic search. The Bin Packing problem is an optimisation problem in which items of different sizes must be packed into a finite number of bins. Conduct desk research and further explain your understanding of the problem.

Dataset and the Bins

The dataset (Appendix) consists of 30 blocks with different heights. The values in the dataset are the heights of the blocks. You are provided with a few bins and the height of each bin is 130cm. Assuming all blocks and bins have the same width and length, optimise the arrangement of the blocks in the bins.

Empirical Experiment

1. Define the solution for the problem and design a suitable data structure to represent the solution.
2. Define, analyse, and discuss the fitness function for the problem.
3. Write a pseudocode for **Random Mutation Hill Climbing** (RMHC) method for your work.
4. Discuss the small change strategy for the problem. Evident the small change strategy either in **pseudocode OR a code snippet**.
5. Based on your experiment design, develop a program (C#) for solving the problem using RMHC and test the program. Document your testing evidence in the report.
6. Using the dataset in the **Appendix**, run an empirical experiment for the program developed in point (5).
7. Discuss your solution and findings for the problem in the report.

Grading Criteria

This project will be graded using the criteria given below.

Criteria	**Marks**
C2: An assessment of ability to work out complexity of an algorithm	10
C3: An assessment of understanding of algorithm design	20
C4: Results analysis and discussion	20
C5: An assessment of data structure design	20
C7: Incorporation of formative feedback	30

Appendix

Dataset

(Also available on Blackboard)